An AI-Augmented Patient Explorer Dashboard

Integrating Synthetic Health Data, Interactive Visualization,
and Local Large Language Model Clinical Summarization

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Abstract

This paper describes the design, architecture, and implementation of a web-based patient explorer dashboard built on synthetic health data generated by the Synthea open-source simulation framework. The dashboard integrates a multi-table relational data model with an interactive Plotly Dash front end and a locally-hosted large language model — Ollama running Meta's Llama 3.1 8B — to produce AI-generated physician-style clinical summary notes for each patient record. The result is a demonstration of how classical data engineering, interactive visualization, and modern generative AI can be combined into a coherent, governed, human-overseen analytical system that surfaces patient complexity in a form immediately useful to clinical and market access audiences.

Introduction

Healthcare analytics has traditionally separated data engineering from data presentation, and both from clinical interpretation. Data engineers build pipelines and schemas; analysts build dashboards; clinicians read reports. Each handoff introduces latency, loss of context, and the risk that the analytical product does not reflect the clinical reality it is supposed to represent.

The Synthea Patient Explorer Dashboard collapses these three functions into a single integrated application. It loads a richly structured synthetic patient dataset, renders it through an interactive visual interface that allows filtering and drill-down by any demographic or clinical characteristic, and then calls a locally-hosted large language model to generate a comprehensive physician-style clinical note for each selected patient — automatically, in real time, from the underlying data.

The project is significant not for novelty in any individual component, but for the discipline of its integration. Each layer — data model, visualization, AI summarization — is transparent, auditable, and under human control. No component is a black box. The AI generates a narrative; the underlying tables remain visible and searchable. The model does not make decisions; it translates structured data into clinically fluent prose that a physician or market access professional can immediately read and evaluate.

Data Foundation: The Synthea Synthetic Health Record

What Is Synthea?

Synthea is an open-source synthetic patient population simulator developed by the MITRE Corporation. It generates realistic but entirely fictitious patient health records by simulating the progression of diseases, clinical encounters, prescriptions, procedures, and demographic characteristics over a patient's simulated lifetime. Records are output in standard formats including HL7 FHIR and flat CSV files.

Synthea's output is not random noise — it reflects epidemiological prevalence rates, clinical care pathways, drug interactions, and demographic distributions calibrated to real-world U.S. health data. A Synthea patient with a seizure disorder will have the appropriate medications, follow-up encounters, EEG procedures, and care plans that a real patient with that condition would accumulate. This makes Synthea an ideal substrate for developing and demonstrating analytics tools without exposing Protected Health Information (PHI).

The Data Model

The Synthea CSV output comprises 18 interrelated files. Figure 1 shows the complete entity-relationship schema, visualized as a directed graph with color-coded node categories representing the functional role of each table.

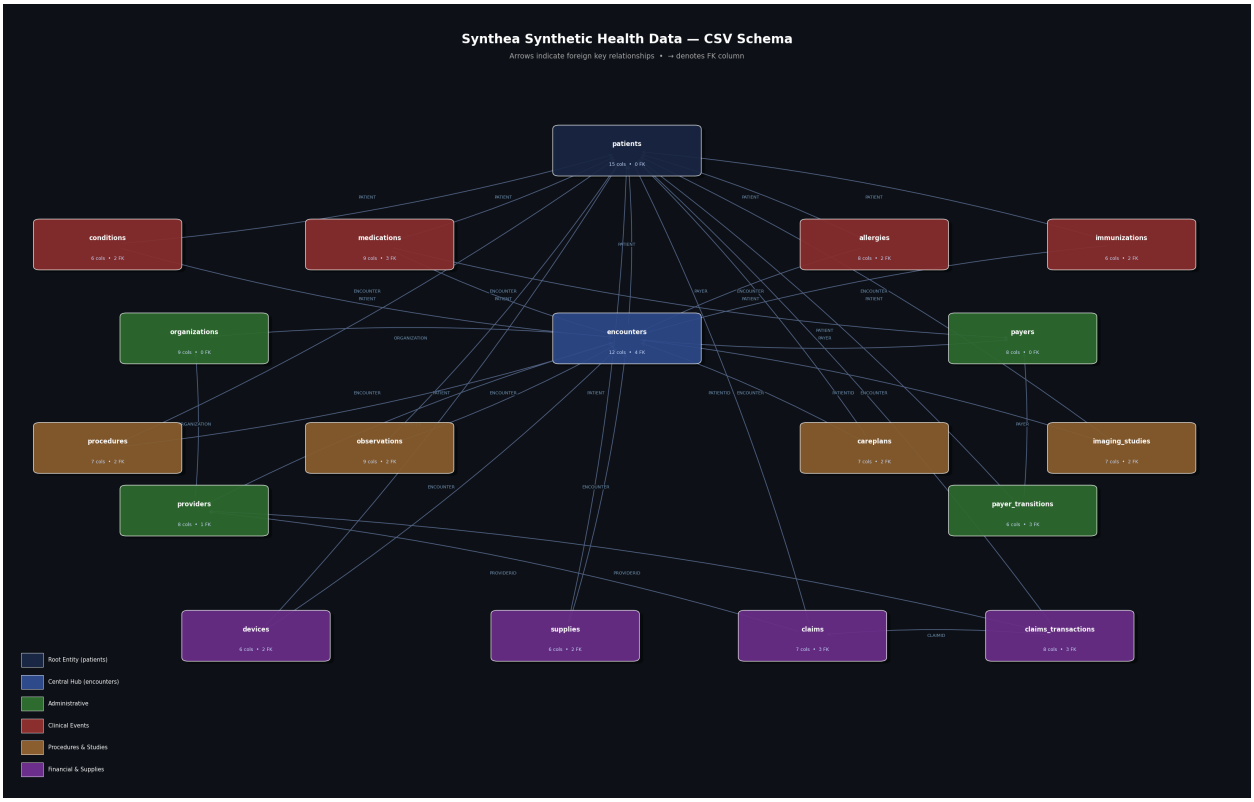


Figure 1: Entity-relationship diagram of the Synthea CSV data model. Nodes are color-coded by role: root entity (patients), central hub (encounters), administrative tables (organizations, providers, payers), clinical event tables (conditions, medications, allergies, immunizations), procedure and study tables, and financial and supply records. Arrows indicate foreign key relationships.

The architecture is hub-and-spoke. The `patients` table is the root entity — every other table references it via a `PATIENT` foreign key. The `encounters` table is the secondary hub: most clinical event tables (conditions, medications, procedures, observations, allergies, immunizations, careplans, devices, supplies, and imaging studies) reference both `patients` and `encounters`, allowing every clinical event to be placed in the context of a specific clinical interaction. Table 1 summarizes the primary tables and their contents.

Table 1: Primary Synthea CSV tables and their analytical role.

Table	Role	Key Contents
`patients`	Root entity	Demographics, address, income, insurance coverage
`encounters`	Central hub	Visit type, provider, payer, costs, dates
`conditions`	Clinical	SNOMED-CT diagnoses with start and stop dates
`medications`	Clinical	Drug name, payer, reason, start and stop dates
`observations`	Clinical	Vitals, labs, survey results, social history
`procedures`	Clinical	CPT/SNOMED procedures with costs and reasons
`allergies`	Clinical	Allergen, type, category, severity
`immunizations`	Clinical	Vaccine description and administration date
`claims`	Financial	Billing claims linked to encounters and providers
`claims_transactions`	Financial	Line-level claim transactions and payments

The dataset used in this demonstration contains 120 synthetic patients located in Massachusetts, generating approximately 7,000 encounters, 4,400 conditions, 5,600 medication records, 93,000 observations, and 20,000 procedures — a substantial analytical corpus despite the modest patient count.

Data Engineering Considerations

Several data quality issues required resolution at load time. The `VALUE` column in several tables stored large integers as comma-formatted strings (e.g., `"104,862"`), which caused numeric conversion to fail silently and return null values downstream. This was corrected at load time with a string replacement prior to coercion. Age was computed from `BIRTHDATE` and `DEATHDATE` (or the current date for living patients). A derived `STATUS` field flagged deceased patients for visual differentiation in the interface. These corrections illustrate a broader principle: even well-structured synthetic data requires validation and normalization before it can be reliably analyzed.

Dashboard Architecture and User Interface

Technology Stack

The dashboard is implemented as a single Python script requiring no database infrastructure, cloud services, or complex deployment. It runs entirely from the directory containing the Synthea CSV files and is portable to any environment with Python and Ollama installed. The primary libraries are:

- **Plotly Dash** — reactive web application framework providing callback architecture, HTML component layout, and Plotly figure integration
- **Plotly Express** — declarative charting for bar charts, timeline (Gantt) charts, and multi-panel line charts
- **pandas** — data loading, filtering, merging, and aggregation
- **Ollama** — local LLM inference server running Meta Llama 3.1 8B, called via HTTP REST API at `localhost:11434`

- **Dash DataTable** — interactive tabular display with sorting, pagination, and conditional row formatting

Layout and Navigation

The interface is organized into two primary regions: a fixed left sidebar containing patient filters and a scrollable right panel containing patient detail. Figure 2 shows the complete dashboard for a selected patient.

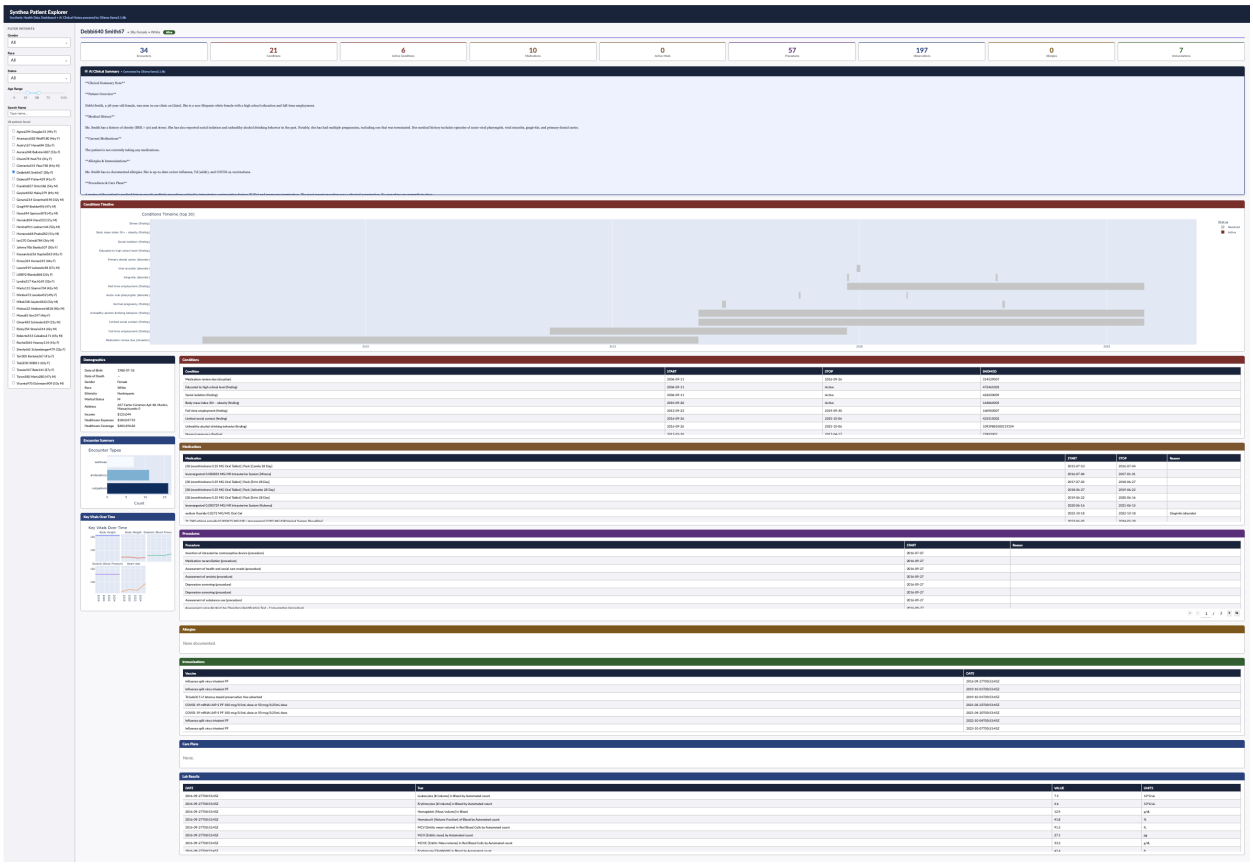


Figure 2: The Synthea Patient Explorer Dashboard showing a selected patient's complete clinical record. The panel includes, from top to bottom: patient identity banner with status indicator, summary statistic boxes, AI-generated clinical note, conditions timeline, and detailed clinical tables. The left sidebar provides demographic and clinical filters with a dynamically updated patient list.

Patient Filtering

The left sidebar provides five simultaneous filter dimensions: gender, race, alive or deceased status, age range via a dual-handle slider, and a debounced name search. All filters are applied through a single Dash callback that re-renders the patient list and auto-selects the first matching patient. The patient count updates dynamically as filters change, giving the user immediate feedback on the size of the resulting cohort. This filtering layer allows the dashboard to function as a population-level exploration tool as well as a patient-level record viewer.

Patient Detail Panel

Selecting a patient triggers the primary rendering callback, which assembles the complete patient view in a single pass. The panel is organized vertically through several functional zones.

Identity Banner

The patient's full name, age, sex, race, and alive or deceased status are presented in a header strip. Deceased patients receive a visually distinct status indicator. This immediately signals clinical context before any detailed data is reviewed.

Summary Statistics

Nine at-a-glance statistic boxes present counts for encounters, total conditions, active conditions, total medications, active medications, procedures, observations, allergies, and immunizations. The distinction between total and active counts for conditions and medications is clinically significant: it separates the patient's current burden from their full historical record.

Conditions Timeline

A Plotly timeline chart renders each condition as a horizontal bar spanning from its start date to its stop date, or to the present if still active. Active and resolved conditions are visually differentiated. This chart communicates the patient's clinical trajectory and the chronological relationship between diagnoses in a way that tabular data cannot.

Clinical Tables

Seven structured tables present the complete record for conditions, medications, procedures, allergies, immunizations, care plans, and laboratory results. Each table scrolls within a fixed-height container and includes alternating row shading. Conditions and medications tables include an active or resolved indicator derived from the presence or absence of a stop date.

Visualization Charts

An encounter type bar chart summarizes the distribution of wellness, emergency, ambulatory, and other visit types. A multi-panel vitals chart displays key measurements — height, weight, heart rate, blood pressure, and BMI — as time series, allowing clinical trends to be identified at a glance.

AI Clinical Summarization

Architecture

The AI summarization layer uses Ollama, an open-source local inference server, running Meta's Llama 3.1 8B parameter model. The model runs entirely on the local machine. No data leaves the system, no external API keys are required, and the system can operate within an air-gapped or HIPAA-compliant network environment without modification.

When a patient is selected, the dashboard constructs a structured prompt from the patient's complete data record and submits it to Ollama's REST API. The response is rendered in the dashboard within approximately 5 to 15 seconds depending on hardware. The AI note panel is positioned immediately below the summary statistics and above the detailed clinical tables — placing the narrative synthesis at the top of the clinical detail, where a physician's summary note would naturally appear in a paper chart.

Prompt Engineering

The prompt is constructed programmatically from the patient's data, organized into labeled sections that mirror clinical documentation practice. The following excerpt illustrates the prompt structure:

```
You are a physician writing a comprehensive clinical summary note.
Write in a professional medical narrative style  clear, concise,
and clinically precise. Structure the note with these sections:
Patient Overview, Medical History, Current Medications,
Allergies & Immunizations, Procedures & Care Plans,
Recent Vitals, Recent Laboratory Results, and Clinical Impression.

ACTIVE CONDITIONS:
  - Seizure disorder (disorder)
  - History of seizure (situation)

CURRENT MEDICATIONS:
  - carbamazepine 20 MG/ML Oral Suspension [Tegretol]

RECENT VITALS:
  Body Weight: 72.3 kg
  Systolic Blood Pressure: 118 mmHg
```

Active and resolved conditions are passed separately. Current and past medications are separated. The most recent unique value for each vital sign and laboratory test is included. Temperature is set to 0.3 to minimize hallucination while preserving clinical fluency. Output is capped at 1,200 tokens, sufficient for a complete structured note without verbosity.

Governance and Human Oversight

The AI note panel is visually distinct from the underlying data tables. It is explicitly labeled as AI-generated, rendered in a serif font with a contrasting border, and positioned so that the source data tables remain fully visible and searchable below it. This design is intentional. It communicates to the user that the note is a synthesis of the underlying data, not a primary source, and that any claim in the narrative can be verified against the structured tables within seconds.

This is the governance principle that distinguishes responsible AI augmentation from black-box automation. The model does not flag anomalies, score risk, or recommend treatment. It translates structured records into fluent prose — a task that is well-bounded, beneficial, and verifiable. The human remains the decision-maker throughout.

Discussion and Conclusion

What This Demonstrates

The Synthea Patient Explorer demonstrates three things simultaneously.

First, that a richly structured multi-table healthcare dataset can be made accessible to non-technical users through a well-designed interactive interface without any database infrastructure, cloud services, or complex deployment. The entire system runs from a directory of CSV files and a single Python script.

Second, that AI-generated clinical narrative can be grounded in structured data rather than generated from model weights alone. The LLM in this system has no independent knowledge of the patient — it knows only what the prompt provides. This substantially reduces the hallucination risk for patient-specific facts while preserving the model's ability to produce fluent, clinically appropriate prose.

Third, that human oversight is not in tension with AI capability. It is the condition under which AI capability is trustworthy. The visible source tables, the explicit AI attribution, and the deterministic data pipeline give the user what they need to evaluate the AI's output critically rather than accepting it uncritically.

Applications in Healthcare Analytics

While the current implementation uses synthetic data, the architecture is directly applicable to real-world use cases. The data model is standard FHIR-compatible — any system that exports conditions, medications, encounters, and observations in a compatible schema can be substituted for the Synthea CSVs. The Ollama integration is model-agnostic and can be replaced with any locally-hosted or enterprise-approved model.

Specific application areas include pharmaceutical market access patient profiling, population health cohort characterization, clinical documentation support, and medical education using realistic synthetic cases. In each context, the same architectural principle applies: AI as translator of structured data into human understanding, with the source data always visible and the human always in control.

Limitations

The current implementation generates AI notes synchronously, causing a pause of several seconds on patient selection. For production use this should be made asynchronous with a loading indicator. The Synthea data, while clinically realistic, does not capture the full complexity of real patient records — particularly the narrative context of physician notes, the nuance of medication dosing changes, or the results of imaging and pathology interpretation. The LLM output, while generally fluent and well-structured, should be reviewed by a qualified clinician before any use in patient care contexts.

Conclusion

The Synthea Patient Explorer is a working demonstration of the principle that AI is most valuable when it augments human understanding rather than replacing human judgment. By grounding the model's output in structured, visible, verifiable source data, and by keeping the underlying tables accessible alongside the AI narrative, the system gives the user both the efficiency of automated synthesis and the confidence of direct data access.

The technical implementation is deliberately simple — a single Python script, local CSV files, and a locally-hosted model. This simplicity is a feature. It demonstrates that sophisticated AI-augmented analytics does not require complex cloud infrastructure, proprietary APIs, or large engineering teams. It requires clear thinking about data architecture, user experience, and the appropriate scope of AI responsibility. That is the pattern that should be built toward across all healthcare analytics work: not AI as a demonstration of capability, but AI as infrastructure — transparent, governed, and woven into the analytical process in a way that makes analysts more capable without making them less accountable.

Prepared by Molly Maskrey, Lead Data Scientist, Forte Analytics. Synthea is an open-source project developed by MITRE Corporation. Llama 3.1 is developed by Meta AI. Ollama is an open-source local inference server. All patient data used in this project is entirely synthetic and contains no Protected Health Information.